

Case Study #1: The Butterfly Ballot and Mistaken Votes in the 2000 Presidential Election

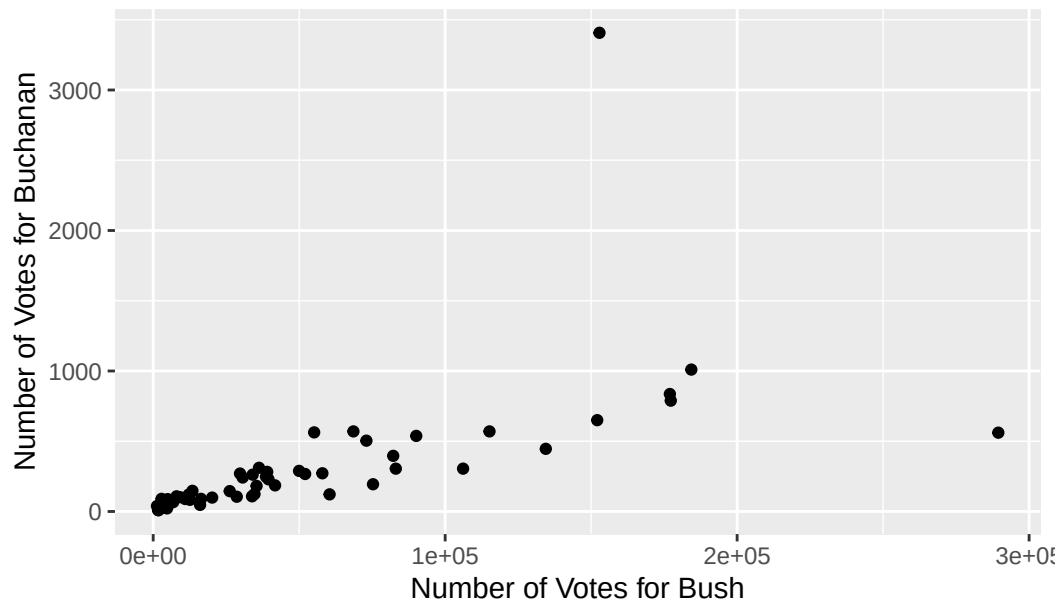
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Introduction

We are investigating the US presidential election that occurred in Florida in the 2000 presidential election. The main candidates were George Bush and Al Gore, but there were also third party candidates such as Pat Buchanan. Many people in Palm Beach County claimed that the unusual shape of the election ballot caused them to mistaken votes for Buchanan rather than Gore in the election. We are conducting statistical analysis to determine the likely number of mistaken votes in Palm Beach County and will be using a linear regression model to predict the number of Buchanan votes relative to the Bush votes across the Florida counties. Here, the response variable is the number of Buchanan votes and the explanatory variable is the number of Bush votes.

Votes for Buchanan and Bush in all 67 counties of Florida

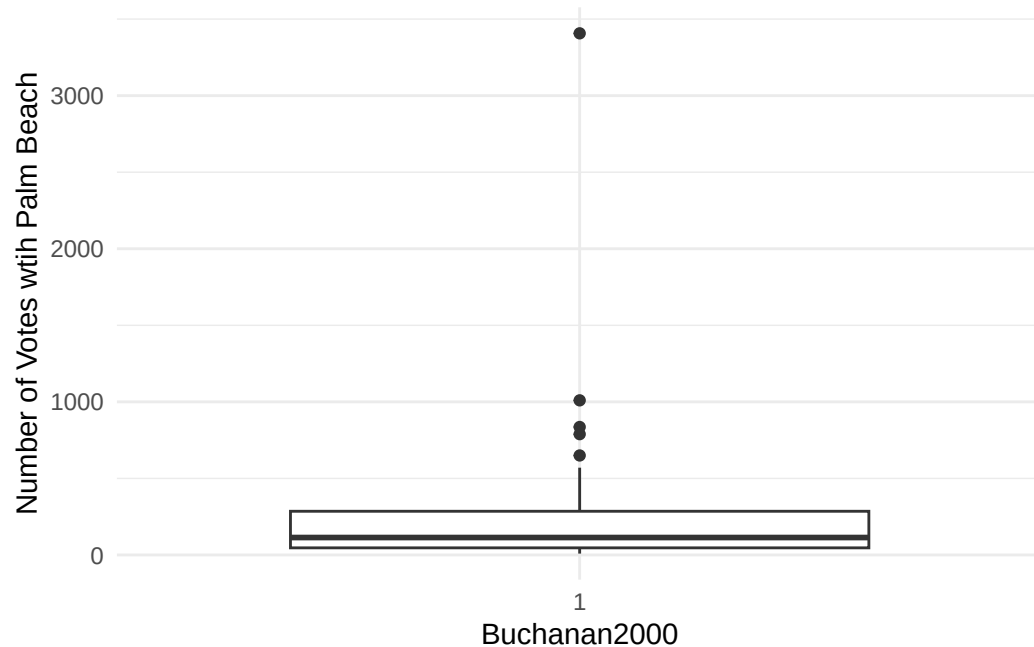


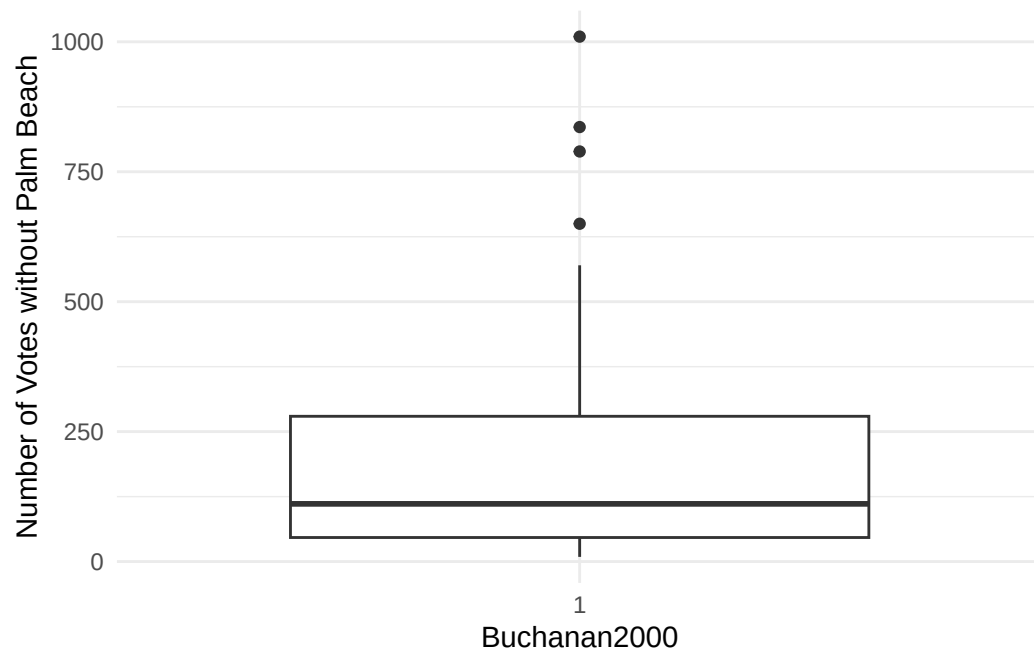
This scatter plot visualizes the number of votes for Bush on the x-axis and number of votes for Buchanan on the y-axis. The data point that appears to be an outlier (over 3000 votes for Buchanan) is that of Palm Beach County.

Exploratory Data Analysis

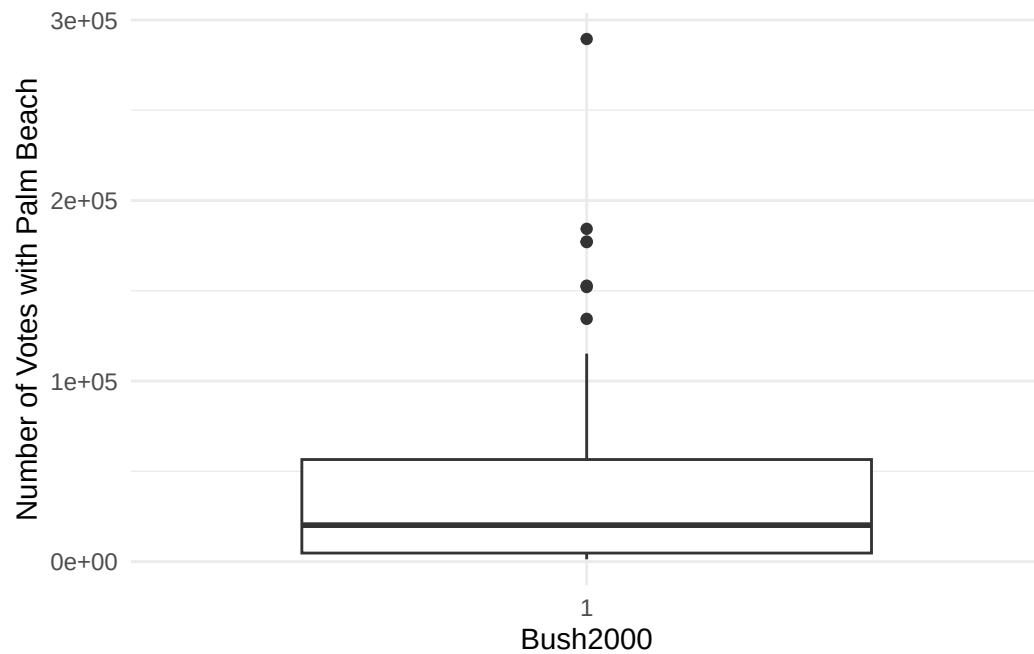
This data set comes from a data problem in *The Statistical Sleuth: A Course in Methods of Data Analysis (2nd ed)*, written by Ramey, F.L. and Schafer, D.W. in 2002. When looking at our original scatter plot, we indeed find that the observation for Palm Beach is a noticeable outlier given the general trend of all other observations. There are a total of 67 counties in Florida, but for the linear regression model, the data set was reduced to 66 counties by removing Palm Beach County, as we suspect that it might be an outlier.

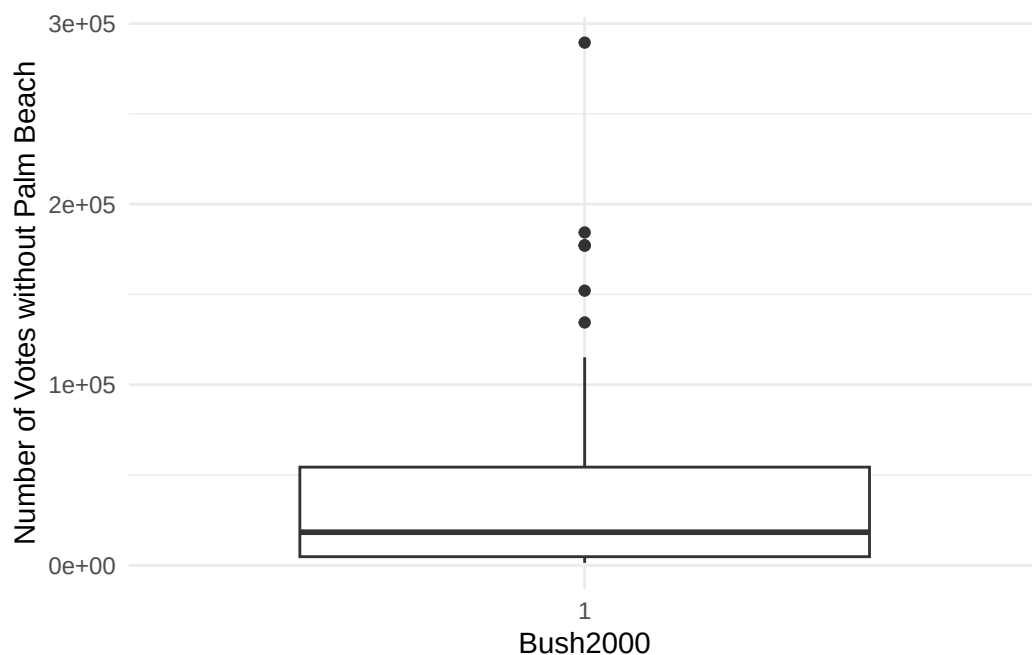
In order to further understand the impact of Palm Beach in the dataset, we take a look at the summary statistics for all votes for Buchanan and all votes for Bush with and without Palm Beach in the form of boxplots.





For Buchanan's votes, the boxplot changes drastically in that the range decreases significantly without Palm Beach vote count in the data.





Bush's boxplot, however, does not seem to change and the range and variance looks to be about the same before and after removing Palm Beach data. This observation is further supported by our summary statistic tables.

n	median	mean	sd
67	114	258.4627	449.3096
66	111	210.7576	223.9688

In comparing the median, mean, and standard deviation of Buchanan's votes before and after removing Palm Beach data, signified by the 'n' column 67 (with Palm Beach) and 66 (without Palm Beach), we see that the standard deviation is halved and the mean decreases by around 50 votes while the median stays around the same value.

n	median	mean	sd
67	20196	43355.76	56989.42
66	18300	41696.82	55772.15

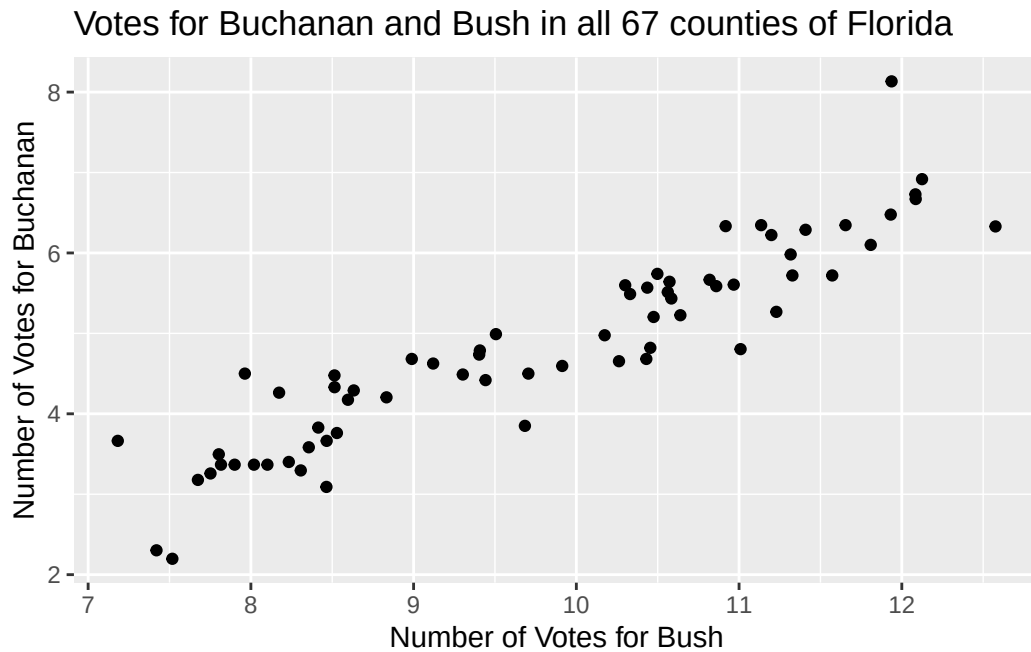
As for Bush, it does not seem that the removal of Palm Beach data impacts the median, mean, and standard deviation as drastically, but as Bush's numbers are so comparatively large, it is important to examine proportional changes rather than simple numerical differences. The table below shows the proportional changes for Bush before and after Palm Beach and for Buchanan before and after Palm Beach, with values demonstrating statistics before the removal of Palm Beach data divided by the statistics after removing Palm Beach data.

candidate	sd_ratio	median_ratio	mean_ratio
Buchanan	2.006126	1.027027	1.226351
Bush	1.021826	1.103607	1.039786

The observations we drew from the boxplots are clearly supported by our proportional changes summary statistics in this table. Palm Beach votes for Buchanan were significantly unusual in that the standard deviation was two times as large and the mean value was 22% as large with the Palm Beach data. For Bush, however, the inclusion of Palm Beach votes manifested in a 2% increase in standard deviation and a 3% increase in mean.

Transforming for Data to fit the Linear Regression Model and Results

To begin, we fit a simple linear regression to the non-transformed variables to predict the number of Buchanan votes from the number of Bush votes across Florida counties. Using a Q-Q plot and fitted residual plot, we found that the normality and linearity conditions were sound. However the equal variance condition was violated, which was indicated by a funnel shape in the observations in which variance increased as the fitted value increased. To overcome this condition violation, we transform our plot by taking the natural log of each variable.



This scatter plot shows the log number of Bush votes on the horizontal axis and the log number of Buchanan votes on the vertical axis.

When we applied a log log transform on the data, we observed that the fitted residual plot demonstrated a more appropriate variance to satisfy the condition. Let x represent the number of Bush votes and $\hat{y}$ be the estimated number of Buchanan votes. We find that:

$$E[\log(\hat{y}) | \log(x)] = 0.731 \log(x) + -2.34.$$

The slope tells us that for a two-fold change in the number of Bush votes, there will be a $2^{0.731}$ multiplicative change in the number of Buchanan votes. The estimates for the model parameters and their standard errors can be found below.

	Estimate	Std. Error	T value	p-value
(Intercept)	-2.34	0.35	-6.61	0
log(Bush2000)	0.73	0.04	20.32	0

Prediction

We predicted the number of Buchanan votes for a county with the same number of Bush votes as Palm beach county - the number of Bush votes was 152846. Because the predicted values are all in log space, we have exponentiated the results and written all values in linear space. The fitted number of votes for Palm Beach county for Buchanan was 589.92. We also constructed a 95% prediction interval about the predicted value, (249.63, 1394.09). There were 3407 Buchanan votes in Palm Beach County, which falls outside of the prediction interval, indicating that this is a highly unusual number of votes for Buchanan. The residual for this estimate is the $y - \hat{y} = 3407 - 589.92 = 2817.08$. We can interpret the residual as the number of mistake votes for the election - based on our model, it is likely that 2817.08 are mistake votes.

Conclusion

The purpose of our case study was to identify whether the votes for Buchanan in Palm Beach County were unusual in the 2000 Presidential Election. We found that the number of votes that Buchanan received in Palm Beach county was indeed very unusual, as the number of votes were far outside of our 95% confidence interval. We found that there were an estimated 2817.08 mistake votes in the Palm Beach election. We identified one limitation of ours to be the lack of prior information on the political alignments per county. While the proportion of Buchanan votes to Bush votes in Palm Beach County were unusual compared to all other counties of Florida, the votes may not have been unusual for the town itself. If we were to be provided historical political data on Palm Beach County, this data may not have been so unusual; the county may simply have population of disproportionately large reform party voters. Given our current information, however, we find that there were indeed an unusual number of votes for Buchanan in Palm Beach, likely due to the confusing layout of the butterfly ballot used in the county.

R Appendix

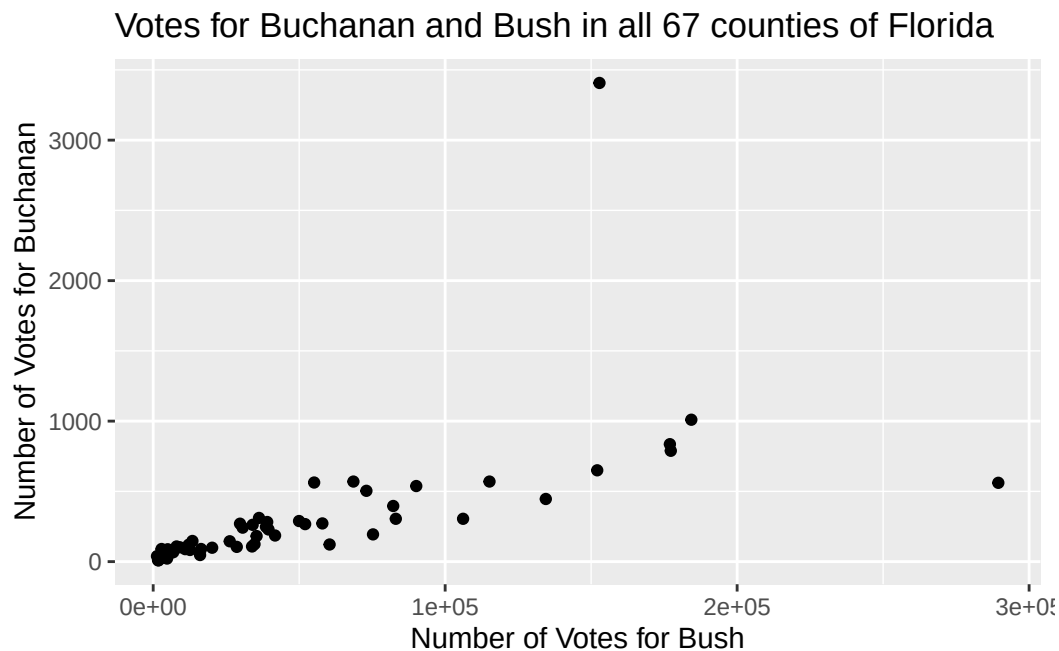
```
# Loading necessary packages
library(tidyverse)
library(Sleuth2)      # the package containing the data for the case study
library(kableExtra)   # for creating nicely formatted tables in Quarto
library(performance)
library(see)
```

```
library(broom)

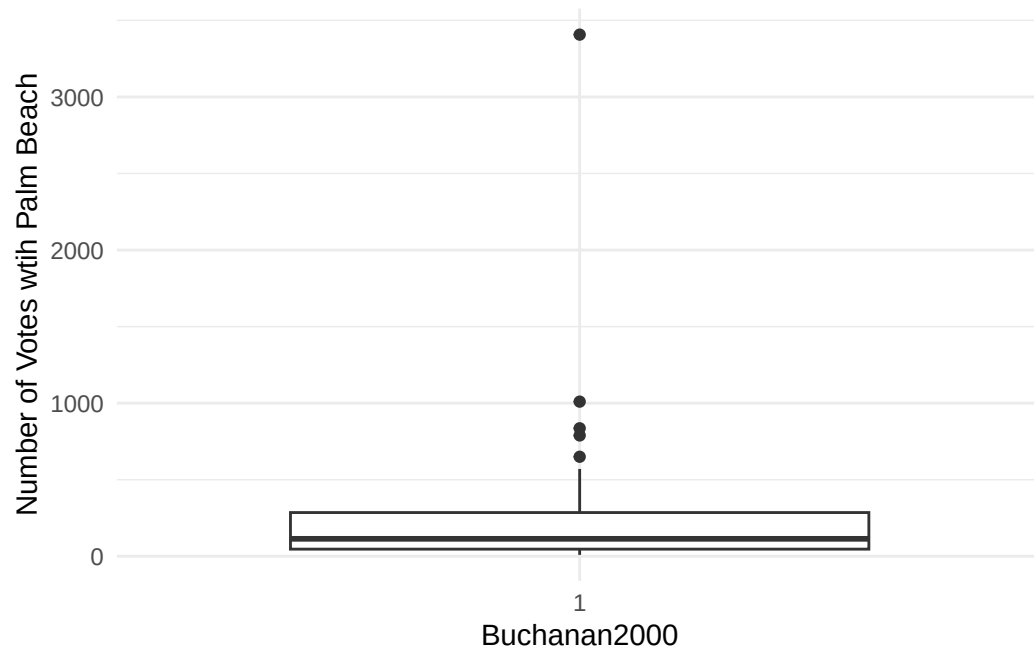
# Loading the case study data
election <- Sleuth2::ex0825

# Creating a second dataset with Palm Beach County excluded
election_wo_pb <- election |> filter(County != "Palm Beach")

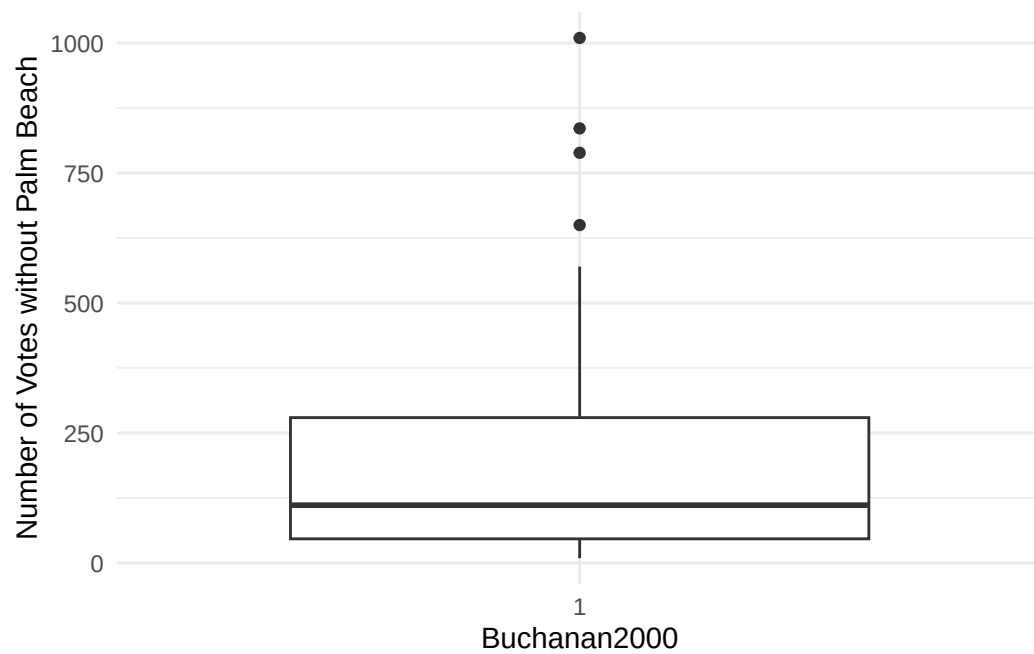
#The scatterplot of the unedited election data
election |> ggplot(aes(x = Bush2000, y = Buchanan2000)) + geom_point() +
  labs(x = "Number of Votes for Bush", y = "Number of Votes for Buchanan", title
    ↪ = "Votes for Buchanan and Bush in all 67 counties of Florida")
```



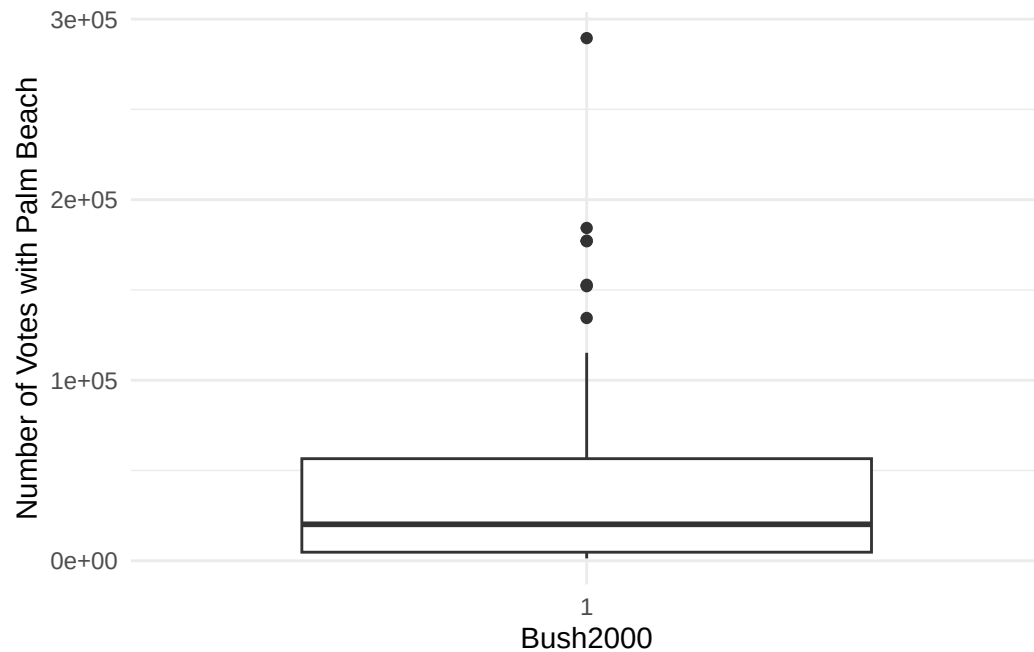
```
#boxplot for buchanan with palm beach
ggplot(election, aes(x = factor(1), y = Buchanan2000)) +
  geom_boxplot() +
  labs(x = "Buchanan2000", y = "Number of Votes wtih Palm Beach") +
  theme_minimal()
```



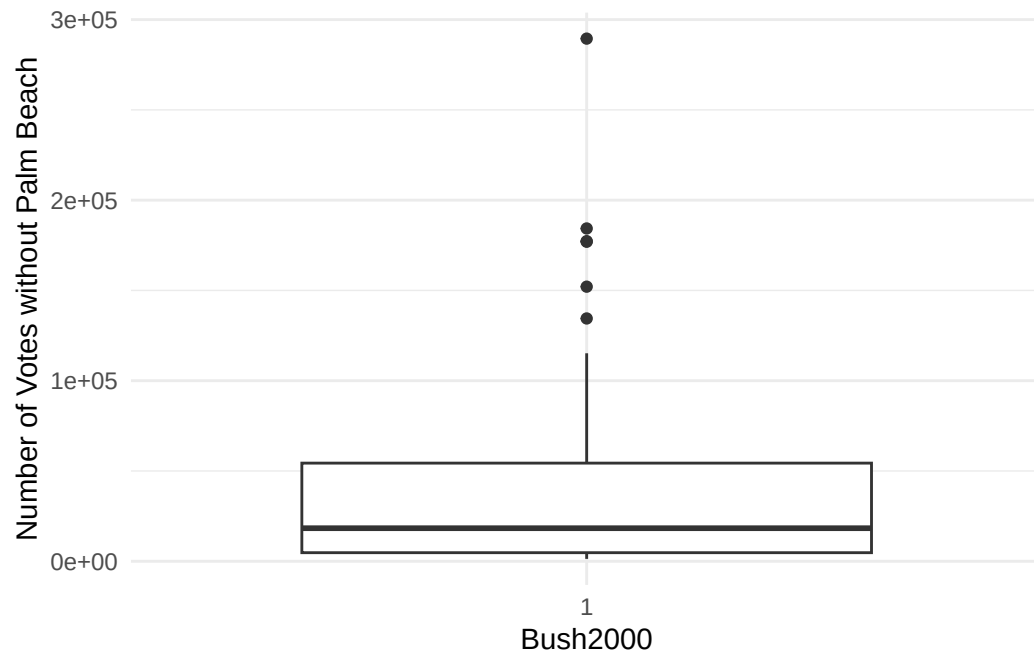
```
#boxplot for buchanan without palm beach
ggplot(election_wo_pb, aes(x = factor(1), y = Buchanan2000)) +
  geom_boxplot() +
  labs(x = "Buchanan2000", y = "Number of Votes without Palm Beach") +
  theme_minimal()
```




```
#boxplot for bush with palm beach
ggplot(election, aes(x = factor(1), y = Bush2000)) +
  geom_boxplot() +
  labs(x = "Bush2000", y = "Number of Votes with Palm Beach") +
  theme_minimal()
```



```
#boxplot for bush without palm beach
ggplot(election_wo_pb, aes(x = factor(1), y = Bush2000)) +
  geom_boxplot() +
  labs(x = "Bush2000", y = "Number of Votes without Palm Beach") +
  theme_minimal()
```



```
# creating a table for the descriptive statistics for buchanan votes with palm
↪ beach
buchan <- election %>%
  summarize(n = n(),
            median = median(Buchanan2000),
            mean = mean(Buchanan2000),
            sd = sd(Buchanan2000))

# creating a table for the descriptive statistics for buchanan votes without palm
↪ beach
buchan_npb <- election_wo_pb %>%
  summarize(n = n(),
            median = median(Buchanan2000),
            mean = mean(Buchanan2000),
            sd = sd(Buchanan2000))

# joining the tables of descriptive statistics for buchanan votes
buchan_table <- buchan %>%
  full_join(buchan_npb, by = c("n", "median", "mean", "sd"))

# mean changes
# 258.4627/210.7576 = 1.226351
# sd is double the original with pb
# 449.3096/223.9688 = 2.006126
# median
# 114 /111 = 1.027027

# creating a table for the descriptive statistics for bush votes with palm beach
```

```

bush <- election %>%
  summarize(n = n(),
            median = median(Bush2000),
            mean = mean(Bush2000),
            sd = sd(Bush2000))

# creating a table for the descriptive statistics for bush votes without palm
  ↪ beach
bush_npb <- election_wo_pb %>%
  summarize(n = n(),
            median = median(Bush2000),
            mean = mean(Bush2000),
            sd = sd(Bush2000))

# joining the tables of descriptive statistics for bush votes
bush_table <- bush %>%
  full_join(bush_npb, by = c("n", "median", "mean", "sd"))

# mean
# 43355.76/41696.82 = 1.039786
# sd is similar for both
# 56989.42/55772.15 = 1.021826
# med
# 20196/18300 = 1.103607

# new table for the ratios of descriptive statistics for data with palm beach
  ↪ divided by data without palm beach
ratio_data <- data.frame(
  candidate = c("Buchanan", "Bush"),
  sd_ratio = c(2.006126, 1.021826),
  median_ratio = c(1.027027, 1.103607),
  mean_ratio = c(1.226351, 1.039786))

#data table with before/after Palm Beach summary statistic for Buchanan votes
kable(buchan_table)

```

	n	median	mean	sd
	67	114	258.4627	449.3096
	66	111	210.7576	223.9688

```

#data table with before/after Palm Beach summary statistics for Bush votes
kable(bush_table)

```

	n	median	mean	sd
67	20196	43355.76	56989.42	
66	18300	41696.82	55772.15	

```
#data table with before/after Palm Beach summary statistic ratios
kable(ratio_data)
```

candidate	sd_ratio	median_ratio	mean_ratio
Buchanan	2.006126	1.027027	1.226351
Bush	1.021826	1.103607	1.039786

```
# Initial linear model before transformations
election_lm_og <- lm(Buchanan2000 ~ Bush2000, data = election_wo_pb)
```

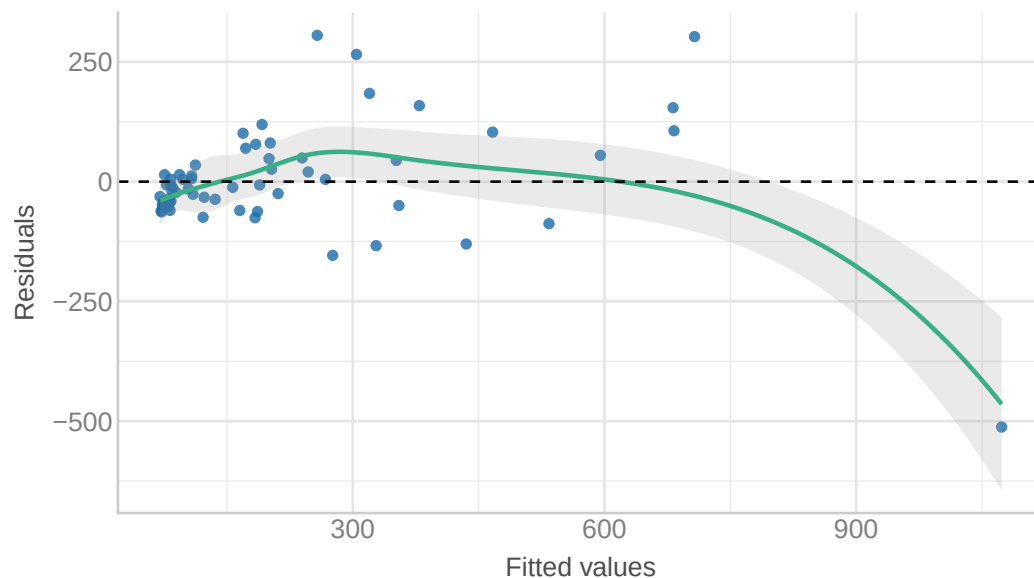
```
# Linearity check for non-transformed model, through which it was determined that
  ↳ the non-transformed model violated the linearity condition
linear_check_og <- check_model(election_lm_og,
                               check = "linearity",
                               panel = FALSE)
```

```
plot(linear_check_og)
```

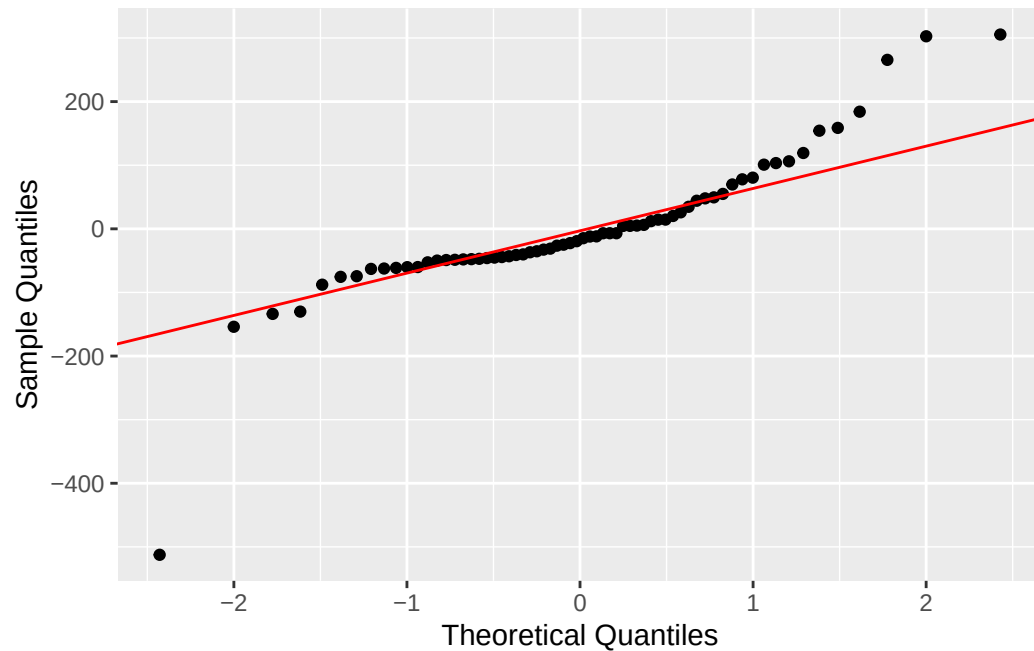
\$NCV

Linearity

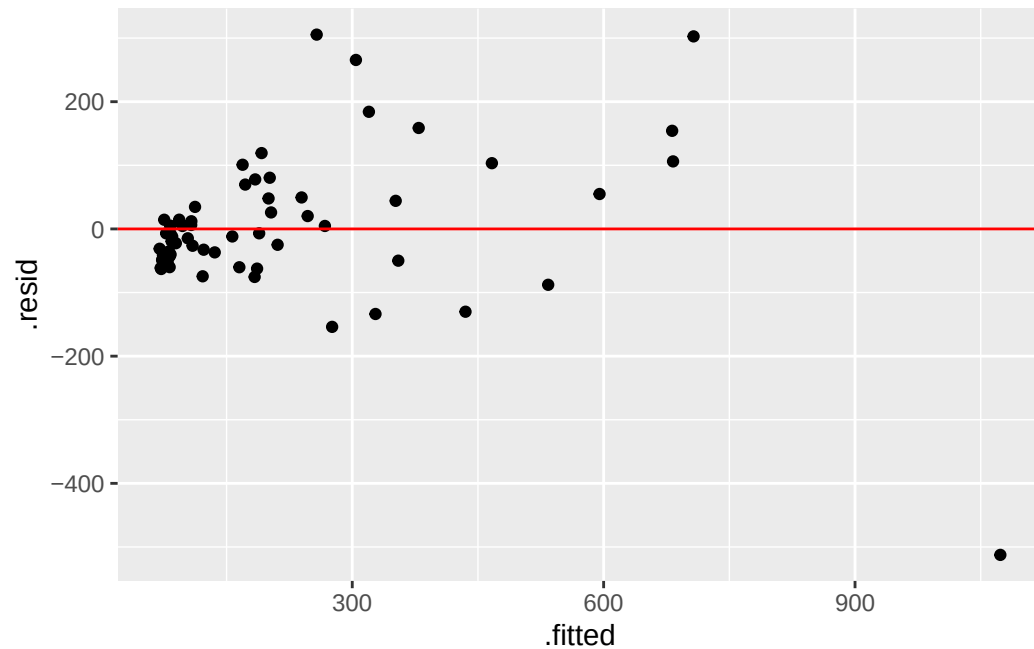
Reference line should be flat and horizontal



```
# Normality of error check for non-transformed model. Shows slight violation of
  ↳ condition, as right tail deviates from exact match between quantiles
election_lm_og |>
  augment() |>
  ggplot(aes(sample = .resid)) +
  geom_qq() + geom_qq_line(col = "red") +
  xlab("Theoretical Quantiles") +
  ylab("Sample Quantiles")
```



```
# Equal variation check for non-transformed model. Severe violation of equal
  ↳ variation, as points show an obvious fan shape
election_lm_og |> augment() |> ggplot(aes(x = .fitted, y = .resid)) +
  ↳ geom_point() + geom_hline(yintercept = 0, col = "red")
```



```
# Fitting a linear regression for the transformed log/log data
election_lm <- lm(log(Buchanan2000) ~ log(Bush2000), data = election_wo_pb)
```

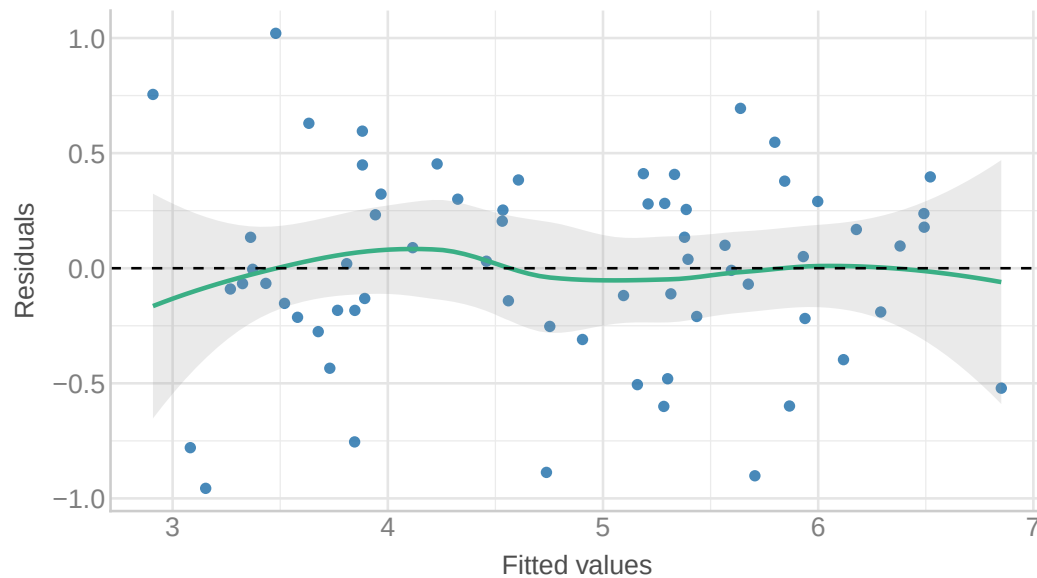
```
# Linearity check for transformed log/log model. Mostly passes linearity check.
linear_check <- check_model(election_lm,
                             check = "linearity",
                             panel = FALSE)

plot(linear_check)
```

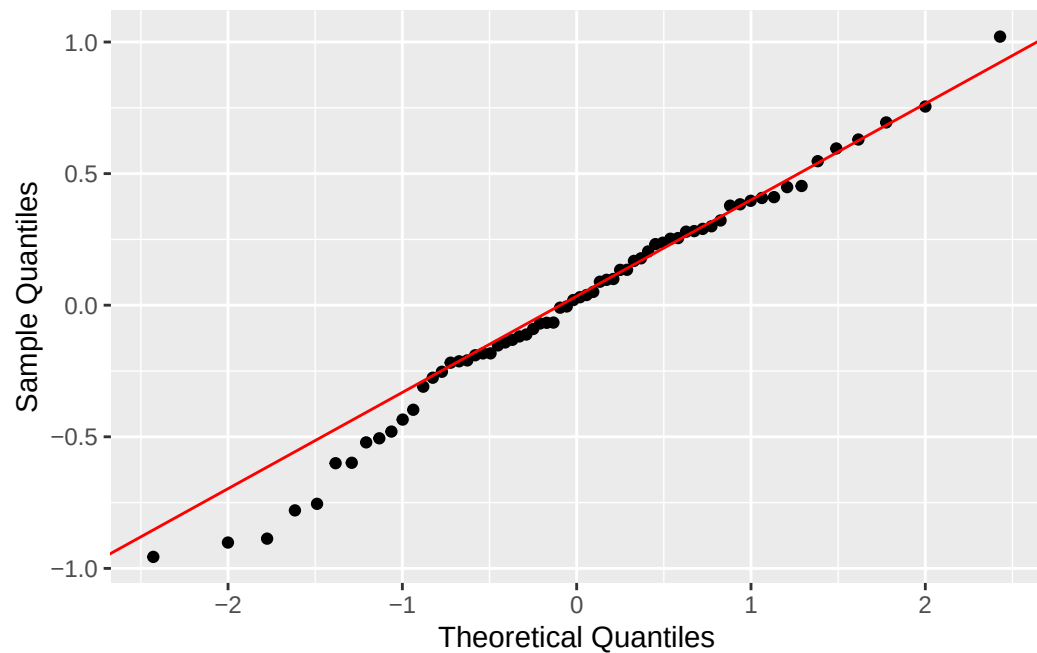
\$NCV

Linearity

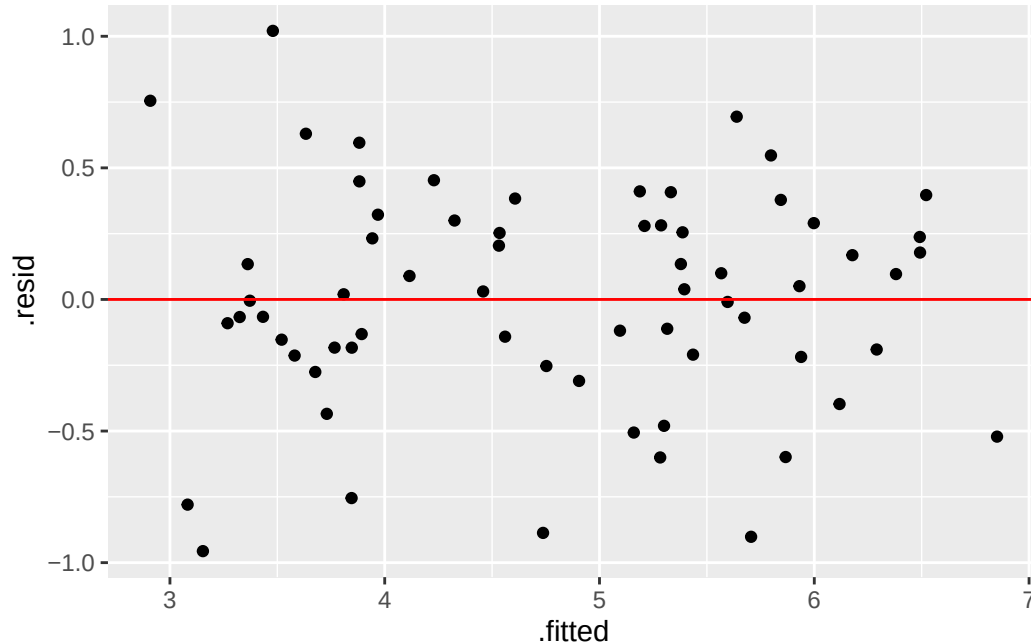
Reference line should be flat and horizontal



```
# Normality of error check for transformed model. Passes - mostly adheres to red  
↪ line  
election_lm |>  
  augment() |>  
  ggplot(aes(sample = .resid)) +  
  geom_qq() + geom_qq_line(col = "red") +  
  xlab("Theoretical Quantiles") +  
  ylab("Sample Quantiles")
```



```
# Equal variance check passed - variance seems constant throughout
election_lm |> augment() |> ggplot(aes(x = .fitted, y = .resid)) + geom_point() +
  ↪ geom_hline(yintercept = 0, col = "red")
```



```
# Representing the regression table as a dataframe (i.e., tidying the summary()
  ↪ output)
election_lm_table <- summary(election_lm)$coefficients
```

```
# Creating a nicely formatted table from the dataframe using the kable package
election_lm_table |> kbl(col.names = c("Estimate", "Std. Error", "T value",
  ↪ "p-value"), align = "c", booktabs = T, linesep="", digits = c(2, 2, 2, 4)) |>
  ↪ kable_classic(full_width = F, latex_options = c("HOLD_position"))
```

	Estimate	Std. Error	T value	p-value
(Intercept)	-2.34	0.35	-6.61	0
log(Bush2000)	0.73	0.04	20.32	0

```
#creating a observation for 152846 Bush votes to find prediction interval
election_int <- data.frame(Bush2000 = 152846)
# Building a 95% confidence interval around mu(2)
election_lm |> augment(newdata = election_int,
  interval = "prediction",
  conf.level = 0.95)
```



```
# A tibble: 1 x 4
  Bush2000 .fitted .lower .upper
  <dbl>    <dbl>  <dbl>  <dbl>
1  152846    6.38   5.52   7.24
```